

A domain prior remembers the site it was written against

[YOUR NAME]

TGDC case study, research track

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Question

A lab that generates enterprise data has to decide whether its generator is any good. The usual check is to train a model on the generated records and score it on held-out real records from the organisation the generator was built for.

When a generator encodes hand-written domain rules, can that check distinguish genuine domain knowledge from a memory of the data those rules were written against?

Hypothesis

H. The conditional structure of real operational data is site-specific rather than domain-general. A generator built from domain priors written against one organisation encodes that organisation. It will work within that site and fail across sites, and evaluation on held-out records from the same site cannot tell the two cases apart.

This matters because a triage agent trained on generated data and validated against the client it was modelled on will be certified as working, then fail at the next client.

The hypothesis was available to fail in two ways. Rules written against one city might carry to another, since municipal service work is broadly similar. Or a prior written with no sight of the data might work in place, in which case the first prior's apparent skill was knowledge and not memory.

Method

Data. Public 311 service request feeds from New York, Chicago and Austin. These are real operational records, written by the organisations that do the work. Each carries an opening time, a service category, an owning department, an intake channel and a closing time. We take two complete calendar weeks from each city, the weeks of 4 May and 8 June 2026.

city	train	test	median resolution
New York	72,206	78,296	5.4h
Chicago	17,291	38,671	118.0h
Austin	5,968	6,029	23.8h

Tickets still open at the pull date are kept and treated as slow, since all are months past any threshold used here. Dropping them removes the slowest work. Chicago closes most of its feed within a second of creation, so those informational records are excluded and only worked tickets remain.

Task. At ticket opening, predict whether it will take longer than the median to resolve. The model sees only what is known at that moment. Logistic regression on one-hot categoricals throughout, so the learner is held constant and differences between arms come from the training data. Recovered skill is the share of the ceiling's advantage over chance that an arm reaches, $(\text{arm} - 0.5)/(\text{ceiling} - 0.5)$.

The three prior conditions. The same procedure, under three information conditions:

1. *Contaminated, in place.* Rules for New York, which we wrote after reading New York's conditional resolution rates, then evaluated on held-out New York tickets. This is the condition a lab is in when it builds a generator by inspecting a client's data.
2. *Carried across.* The same New York rules, unmodified, on Chicago.

3. *Blind, pre-registered.* Rules for Austin, written from Austin’s service type and department vocabulary alone. The taxonomy was pulled with the query restricted to category names and row counts. No duration, rate or per-category outcome was requested or displayed. The rules were then written, committed with a recorded prediction, and only then evaluated.

The pre-registration is checkable. The rules are commit 8973439 and the evaluation is commit 4400226.

Results

A prior works only where it was written

rules written for	evaluated on	seen this city’s rates	AUC	ceiling	recovered
New York	New York	yes	0.840	0.918	81%
New York	Chicago	no	0.496	0.729	−2%
Austin	Austin	no, pre-registered	0.494	0.886	−1%

The contaminated prior recovers 81% of the achievable skill in the city it was written against. The same procedure recovers nothing anywhere else, and nothing at all when the author writes blind.

This is not a coverage failure. The Austin rules fired on 98.9% of tickets, producing scores spread from 0.05 to 0.95. Tickets they called slow ran slow 60.1% of the time and tickets they called fast ran slow 52.9% of the time, a gap of seven points that does not survive as ranking skill.

The carried-across case fails for a visible reason. The department rules match 0.0% of Chicago rows, being New York acronyms. The keyword list is New York housing-stock vocabulary, and in Chicago it points the wrong way: rows it calls slow are slow 47.3% of the time against 58.6% for everything else.

Why the prior fails, in detail

The Austin prior rests on four claims about the nature of municipal work. One person attending once is fast. An inspection that opens a process is slow. Work needing a crew and materials is slow. A fixed-cycle routine service is fast. Each is plausible, and each is wrong somewhere.

service type	predicted slow	actually slow
ARR Compost	5%	96%
Vehicle Abatement Report	5%	94%
ARR Bulk	5%	92%
Animal Protection, Loose Dog	5%	81%
Traffic Signal Maintenance	90%	8%
Parking Violation Enforcement	30%	0%
Request Code Officer	95%	82%

A traffic signal is physically crew work but administratively a same-day safety priority. Compost collection is physically routine, but the ticket stays open across the collection cycle. Vehicle abatement is a police matter subject to a statutory waiting period. In each case the duration is set by the organisation’s workflow and its closing conventions, not by the nature of the job. Reasoning about the work cannot recover that.

Why the in-place check is easy to pass

Two measurements explain why nobody notices. The New York task is close to a lookup on one column: the service category alone scores 0.916 against the 0.918 ceiling, so it carries everything. And the single rule `department is not NYPD` scores 0.850, beating the thirty-keyword prior’s 0.840.

A generator ladder makes the same point. Trained on generated New York records and scored on real ones:

generator	AUC	recovered	records that cannot occur
real records (ceiling)	0.918	100%	0.0%
marginal only	0.466	−8%	70.5%
marginals + pairwise	0.917	100%	0.0%
empirical joint	0.917	100%	0.0%

An evaluation on this task will certify almost anything that gets one column roughly right.

Two predictions that were wrong

Before evaluating Austin we recorded a predicted range of 0.60 to 0.72, on the reasoning that genuine domain knowledge ought to be worth something even if less than a contaminated prior. The observed 0.494 is below that band. The prediction was wrong and the hypothesis holds in a stronger form than expected. The contaminated prior's advantage was not mostly memory, it was all of it.

A second pre-registered hypothesis also failed. We expected generated records violating real-world constraints to explain the loss in downstream skill. Marginal-only generation does invent impossible department and category pairs, 70.5% of its records, and it does lose everything. But the standard bar of marginals plus pairwise invents none at all and loses nothing. That is failure mode one of the three written down in advance: at the fidelity order that matters, there was nothing to measure.

Limits

Three cities, one domain, one task family, one model class. Municipal service records may be unusually institution-bound. A domain with genuinely standardised process, such as a regulated industry or a franchise operation, could behave differently, and that is the first thing we would test next.

We wrote all three priors by one procedure. A different author, or someone with real municipal operations experience, might produce rules that carry. Nothing here separates the procedure from the person.

Chicago's June week ran roughly twice the volume of its May week, so the carried-across arm compares two operating conditions as well as two cities. Its ceiling of 0.729 is also the lowest of the three, leaving less skill available to recover.

We tried a harder framing, predicting speed within a service category to remove the dominant column. Its ceiling is 0.514, too close to chance to separate anything, so we report it as a limit and not a result.

New York carries known artefacts, including one department that closes 89.7% of its tickets at exactly midnight, so its recorded durations are administrative.

That many high-order joints share low-order marginals, and that low-order fidelity therefore fails to imply downstream utility, is established. We include the ladder because it explains why the contamination goes unnoticed.

Finally, the contamination is irreversible. Having inspected New York and Chicago, we can no longer write a blind prior for either. A prior-driven generator can be evaluated honestly once per dataset, and only before anyone has looked.

What it would take to answer the general question

Pre-registration would have to become the default rather than an experiment. Rules written against a taxonomy with outcomes withheld, committed, then scored once, is the only procedure here that produced a number worth trusting. It cost an afternoon.